LabelLite: Label-Efficient, Uncertainty-Aware Image Classification with Frozen Foundation Models

Abstract

Expert labels are expensive in pathology and remote sensing, and a model used there has to be able to flag the predictions it cannot back up. We benchmark four frozen vision backbones (ResNet-50, DINOv2, CLIP, and the pathology model Phikon) with linear probes against a fully fine-tuned ResNet-50 on PatchCamelyon (PCam) and EuroSAT at 1%, 10% and 100% of training labels, and add temperature scaling, split-conformal triage, a synthetic stain shift, and simulated active learning. On PCam, a linear probe on Phikon features trained on 1% of the labels (AUROC 0.980) outperforms the ResNet-50 fine-tuned on all labels (0.966). Conformal triage auto-labels 90.1% of PCam test patches at a 95% coverage target with 4.7% error among them, but coverage falls short by up to 4.7 points because calibration and test slides differ, and further under stain shift. On EuroSAT, uncertainty sampling reaches random sampling's final accuracy with 58–65% fewer labels; on PCam it does not help. This is a research benchmark and makes no clinical claims.

1 Introduction

Labeled data in pathology and remote sensing is expensive because it needs a domain expert. Self-supervised foundation models promise useful features without task labels, and conformal prediction promises a distribution-free way to decide which predictions to trust. We ask five questions on two image domains: whether frozen features plus a linear probe match a fine-tuned CNN at low label budgets (RQ1); whether their confidences are calibrated (RQ2); whether split-conformal triage can auto-label images at a controlled error (RQ3); how these properties hold under a stain shift (RQ4); and whether uncertainty-driven active learning saves labels (RQ5). All code, configurations, split IDs and result tables are released; every number below is produced by the pipeline.

2 Related work

Frozen features and foundation models. Linear probes on frozen features are a standard test of representation quality, and ImageNet accuracy predicts transfer only partly [11]. Self-supervised models such as DINOv2 [13] and language-supervised CLIP [15] are evaluated this way on natural images. In pathology, self-supervised models trained on large slide collections, among them Phikon [5], UNI [4] and CONCH [12], report strong linear-probe results on patch-level pathology benchmarks. We compare general-purpose and pathology models *across* domains, and against fine-tuning at matched label budgets.

Calibration and conformal prediction. Modern networks are often overconfident, and temperature scaling is a simple post-hoc fix [7]. Split conformal prediction [25, 1] turns any classifier into set-valued predictions with marginal coverage under exchangeability. LAC [18] gives the smallest sets on average, while APS [16] and RAPS [2] aim for more adaptive sets. Coverage guarantees break under distribution shift unless it is modeled, for example by weighting for covariate shift [23]. Uncertainty estimates in general degrade under shift [14]. We measure this with fixed thresholds as a deployment would.

Shift in pathology. Stain and scanner differences across hospitals are a major source of shift in histopathology. Stain augmentation and normalization are common mitigations [22]. Camelyon17 [3] and its WILDS version [10] provide real cross-hospital splits. We use a controlled stain perturbation [17] instead (Section 6).

Active learning. Uncertainty sampling [19], Bayesian acquisition [6] and diversity-based core-set selection [21] are the main families. We test the simplest uncertainty and core-set strategies, plus conformal set size, on top of frozen features, where retraining a probe each round is cheap.

3 Method

Data. PCam [24] has 327,680 H&E lymph-node patches (96×96 px, binary tumor label) with official train/valid/test splits from disjoint slides. EuroSAT RGB [9] has 27,000 Sentinel-2 images (64×64 px, 10 land-cover classes) without official splits; we use a fixed stratified 60/10/10/20 train/val/cal/test split. The PCam calibration split is the official valid split (32,768); its test split (32,768) comes from other slides. Label budgets of 1% and 10% are class-stratified subsets of train, drawn with 3 seeds.

Models. Frozen backbones: ResNet-50 [8] (ImageNet-1k), DINOv2 ViT-B/14 [13], CLIP ViT-B/16 [15], and Phikon [5], an iBOT ViT-B/16 pretrained on TCGA (an open substitute for gated pathology models; no Camelyon slides in pretraining). Images are resized to 224×224 and features cached in float16. On L2-normalized features we fit an L2-regularized logistic regression (C chosen on val) and a k-NN classifier ($k = 20$), and evaluate CLIP zero-shot. The baseline is ResNet-50 fully fine-tuned with

Table 1: Test performance (mean ± std over 3 seeds at 1%, 10%; single run at 100%). PCam: AUROC. EuroSAT: accuracy. LP = linear probe, FT = fine-tuned.

PCam AUROC	1%	10%	100%
ResNet-50 LP	.907±.004	.914±.002	.913
ResNet-50 FT	.922±.006	.953±.004	.966
DINOv2 LP	.927±.003	.936±.001	.941
CLIP LP	.900±.002	.906±.001	.910
Phikon LP	**.980**±.001	**.979**±.005	**.982**
EuroSAT accuracy	1%	10%	100%
ResNet-50 LP	.870±.001	.939±.002	.966
ResNet-50 FT	.865±.001	**.968**±.002	**.984**
DINOv2 LP	**.880**±.012	.939±.002	.965
CLIP LP	.862±.007	.930±.003	.958
Phikon LP	.792±.003	.904±.004	.952

AdamW (warm-up, cosine decay), color jitter and early stopping on val, in bf16.

Calibration and conformal triage. The calibration split is halved (stratified): one half fits a temperature [7], the other calibrates split conformal prediction [25, 1] on temperature-scaled probabilities with the LAC score $1 - \hat{p}_y$ [18] and randomized APS [16], at $\alpha \in \{0.1, 0.05\}$. An image whose set has exactly one class is auto-labeled; otherwise it is routed to an expert. We report coverage, mean set size, the auto-labeled share, the error among auto-labeled images, and system error (wrong auto-labels over all images, $\leq \alpha$ if the expert is always right). Coverage is compared with its exact finite-sample distribution under exchangeability (calibration Beta law plus test binomial noise); as a control we also calibrate on a random half of test.

Stain shift. For RQ4, PCam test patches are shifted in Ruifrok–Johnston optical-density stain space [17]: at severity $s \in \{1, 2, 3\}$ hematoxylin is scaled by $1-0.15s$ and eosin by $1 + 0.25s$. These values were fixed by inspecting example images before any shifted results existed. Models, temperatures and conformal thresholds stay fixed from clean data.

Active learning. Starting from the 1% subset, 10 rounds each add 1% of the train pool, chosen by random sampling, least confidence, entropy, LAC conformal set size ($\alpha = 0.1$, ties broken randomly), or k-center greedy [21] on DINOv2 and Phikon features. The oracle is the dataset label; 3 seeds per strategy.

Explanations. Grad-CAM [20] of the predicted class's log-probability, at ResNet-50's `layer4` and at the patch tokens entering each ViT's last block, with the linear probe as a differentiable head. Maps are used qualitatively only.

Table 2: Conformal triage at 10% labels, LAC score (mean over 3 seeds, %). Cov. = coverage; Auto = auto-labeled share; Err = error among auto-labeled; Sys = system error.

Target	Model	Cov.	Auto	Err	Sys
PCam					
95%	Phikon LP	95.8	90.1	4.7	4.2
95%	DINOv2 LP	95.3	73.2	6.4	4.7
95%	ResNet-50 FT	94.4	80.9	6.9	5.6
90%	Phikon LP	90.4	96.8	6.6	6.4
90%	ResNet-50 FT	88.1	98.3	12.1	11.9
EuroSAT					
95%	DINOv2 LP	95.7	95.5	4.2	4.0
95%	ResNet-50 FT	94.6	96.1	1.6	1.5
90%	DINOv2 LP	90.5	93.9	3.6	3.4

4 Results

RQ1: label efficiency. Table 1 and Fig. 1. On PCam, Phikon dominates at every budget, and its 1% probe (accuracy 0.925, AUROC 0.980) beats the ResNet-50 fine-tuned on 100% of labels (0.886, 0.966). At 100% labels the accuracy gap (+0.038) is significant (McNemar, Holm-corrected $p < 0.001$); the paired bootstrap of the AUROC gap (+0.016) gives $p = 0.072$, the smallest value reachable with 1,000 resamples after Holm correction over 36 pairs, so that test is underpowered rather than negative. On EuroSAT, general-purpose features win at 1% (DINOv2 0.880 vs. fine-tuned 0.865), fine-tuning wins from 10% on, and the best probe at 10% reaches 95.5% of the fine-tuned model's full-label accuracy (0.939 vs. 0.984). The pathology model is the weakest probe on satellite images, and general-purpose models are far behind Phikon on pathology: domain-specific pretraining matters in its own domain only. CLIP zero-shot is at chance on both datasets (accuracy 0.488 on PCam, 0.504 on EuroSAT) with our prompts.

RQ2: calibration. Linear probes from 10% labels up have expected calibration error (ECE, 15 bins) of at most 0.035 before any correction. Miscalibration appears in the low-label and fine-tuned regimes: the fine-tuned ResNet-50 on PCam has ECE 0.115 ± 0.014 at 1% labels and 0.056 at 100% (temperatures 2.58 and 2.00, i.e. overconfident), reduced by temperature scaling to 0.028 ± 0.005 and 0.010. On EuroSAT at 1% labels both ResNet-50 variants start near 0.15 and drop to 0.017–0.024. Phikon on PCam is already calibrated (ECE 0.007–0.013) and temperature scaling leaves it at 0.010–0.014.

RQ3: conformal triage. Table 2. With Phikon at a 95% target, 90.1% of PCam test patches receive a single-label set, with 4.7% error among them and 4.2% system error. APS produces larger sets: 73.3% auto-labeled at 3.4% error for the same model. On EuroSAT, coverage lies within 3 standard deviations of its expected value in all 144 (run, score, α) cases. On PCam this holds in only 94 of 144, with deviations from -4.7 to $+1.3$ points; calibrating

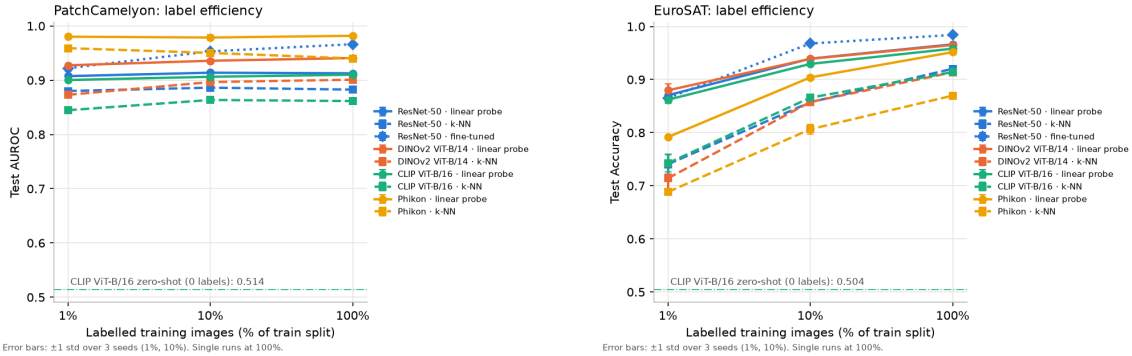


Figure 1: Label efficiency: test AUROC on PCam (left) and accuracy on EuroSAT (right) against the fraction of training labels, mean ± std over seeds.

on a random half of test instead restores 144 of 144. The gap is therefore due to the slide-level difference between PCam's valid and test splits, a mild shift that already violates exchangeability. Class-conditional coverage is uneven: at 10% labels and a 90% target, the least-covered class sits as low as 73.9% (Phikon LAC on EuroSAT) and LAC under-covers tumors on PCam (e.g. 83.6% for Phikon), whereas APS keeps the minimum class coverage closer to target (87.7% and 87.8% for the same settings).

RQ4: stain shift. Fig. 2. From severity 0 to 3 (10% labels, LAC at a 90% target), the fine-tuned ResNet-50 degrades most: AUROC 0.953 → 0.875, coverage 88.1 → 79.4%, and system error 11.9 → 20.6%. Frozen DINOv2 features are the most stable (AUROC 0.936 → 0.928, coverage 89.7 → 88.7%), Phikon remains the most accurate (0.979 → 0.967; coverage 90.4 → 87.2%), and CLIP and ResNet-50 probes lose 4.7 and 3.2 coverage points. APS degrades less than LAC for every frozen probe (e.g. DINOv2 stays at 90.0%). Temperature-scaled ECE stays below 0.02 for DINOv2 and Phikon but reaches 0.064 ± 0.011 for CLIP. Conformal thresholds set on clean data do not carry their guarantee across a stain change, and the size of the failure depends strongly on the representation.

RQ5: active learning. Fig. 3. On EuroSAT, least-confidence sampling ends at 0.959 (DINOv2) and 0.940 (Phikon) accuracy against 0.941 and 0.903 for random sampling at 1,782 labels, and reaches random's final accuracy with 623 and 746 labels: 65% and 58% fewer labels than random itself needs. Entropy is close behind (58%, 55%); conformal set size saves 62% with Phikon but never reaches random with DINOv2. On PCam no strategy beats random: AUROC is flat from the 1% start (Phikon random 0.982), and least confidence (0.975), entropy (0.977) and k-center (0.965) end below it; a plausible but untested explanation is that uncertainty sampling concentrates on ambiguous patches once the classifier has saturated. k-center greedy is the weakest strategy in three of four settings.

5 Failure modes

The dominant PCam error of the reference Phikon probe (10% labels, seed 0) is the costly one: 87.3% of its 2,491 test errors are tumor patches predicted normal. On EuroSAT, the most frequent DINOv2 confusions are between visually similar classes: AnnualCrop and PermanentCrop (55 of 320 errors) and Highway and River (37).

In addition, 100 random errors per dataset were tagged by failure type, from each image and its Grad-CAM map. No pathologist or remote-sensing expert was involved, so the counts describe what a non-expert can see. On PCam, 57 of the 100 showed no tumor evidence a non-expert could recognize; 17 were clear misses with visible clusters of larger cells; 10 had tissue only at the patch edge; 7 had an adipose center, which points to coarse annotation; 7 were torn or blurred; and 2 had unusual staining. EuroSAT errors are easier to read: 62 were look-alike classes, mostly crop and vegetation types, 16 were mixed scenes containing both classes, 13 were plain model errors, 5 had the defining feature in a small corner of the image, 2 looked mislabeled and 2 had haze or a color cast. The Grad-CAM galleries of confident errors and uncertain images are released with the code. The ViT maps are limited to a 14×14 or 16×16 patch grid and should not be read as evidence of what the model relies on.

6 Limitations

PCam's patch-level labels simplify slide-level diagnosis, and EuroSAT images are upsampled from 64×64 px. The stain shift is synthetic; the planned cross-hospital test (Camelyon17-WILDS) was not available for download. Conformal guarantees are marginal and assume exchangeability, which PCam's own splits already violate. The fine-tuned baseline uses one fixed recipe, and fine-tuning was not part of the active-learning study, whose oracle is the dataset label. Three seeds per budget, and pairwise tests on seed 0 only, limit the statistical resolution.

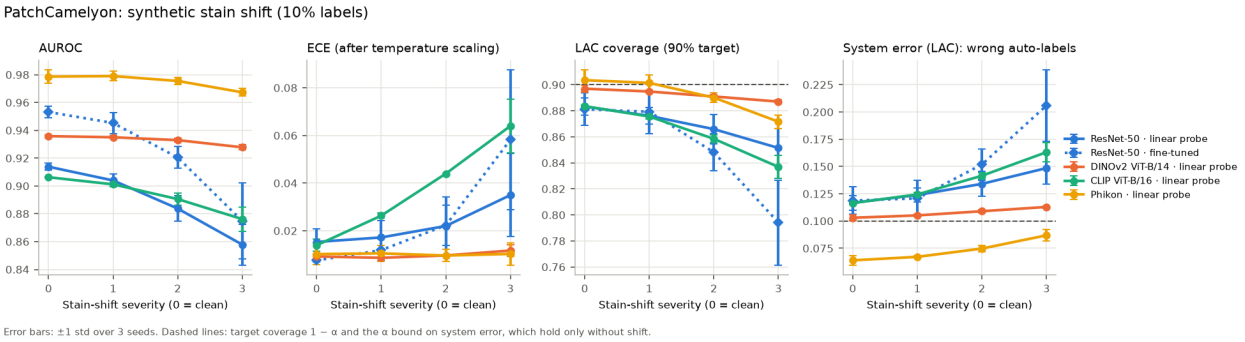


Figure 2: Synthetic stain shift on PCam (10% labels): metrics against severity, with models, temperatures and conformal thresholds fixed from clean images.

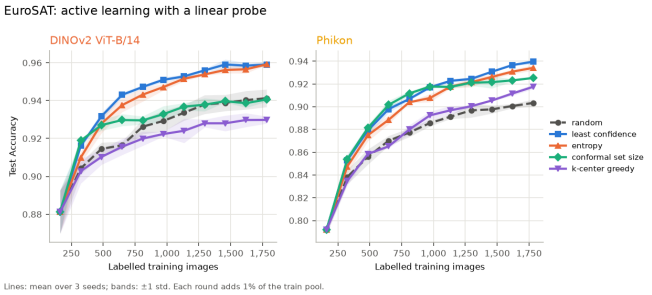


Figure 3: Active learning on EuroSAT: test accuracy against labeled images (mean ± std over 3 seeds), DINOv2 (left) and Phikon (right).

7 Ethics

This is a benchmark on public datasets, not a diagnostic tool, and the pathology results make no clinical claim; a high patch-level AUROC does not establish clinical usefulness, and the model's dominant error, missed tumor, is the clinically costly direction. Coverage is an average over images, not a per-patient guarantee, and fails under shift. Phikon's weights are licensed for non-commercial research; weights and raw images are not redistributed.

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